

analog filter.

Converting Analog Transfer Function, $H_a(s)$ into digital Transfer Function, $H(z)$

Analog Transfer function represented in terms of partial fraction expansion is expressed as :

$$H_a(s) = \frac{A_1}{s-p_1} + \frac{A_2}{s-p_2} + \frac{A_3}{s-p_3} + \dots + \frac{A_M}{s-p_M}$$

In general,

$$H_a(s) = \sum_{i=1}^M \frac{A_i}{s-p_i} \quad \text{--- (1)}$$

Where, $A_1, A_2, A_3, \dots, A_M$ are Coeff. of partial fraction expansion.

$p_i = p_1, p_2, p_3, \dots, p_M$ are poles.

$s \Rightarrow$ Laplace operator.

Then, $h_a(t) \xleftrightarrow{\text{ILT}} H_a(s)$ and is defined as,
 ↖ Inverse Laplace Transform

$$h_a(t) = \sum_{i=1}^M A_i e^{p_i t} \quad \text{--- (2)}$$

Now, The impulse response of the digital filter is obtained uniformly by sampling the impulse response of the analog filter i.e. :

$$h[n] = h_a(nT) \quad \text{--- (3)}$$

Where, $h_a(nT)$ is the analog filter impulse response & T is the sampling period. So,

$$h[n] = \sum_{i=1}^M A_i e^{p_i nT} \quad \text{--- (4)}$$

Taking Z-transform is defined as,

$$H(z) = \sum_{n=0}^{\infty} h[n] z^{-n} \quad \text{--- (5)}$$

⑤

From eqⁿ ④ & eqⁿ ⑤, We get,

$$H(z) = \sum_{n=0}^{\infty} \left[\sum_{i=1}^M A_i e^{p_i n T} \right] z^{-n}$$

$$= \sum_{i=1}^M A_i \sum_{n=0}^{\infty} e^{p_i n T} z^{-n}$$

$$= \sum_{i=1}^M A_i \sum_{n=0}^{\infty} \left(e^{p_i T} z^{-1} \right)^n$$

$$= \sum_{i=1}^M A_i \times \frac{1}{1 - e^{p_i T} z^{-1}}$$

$$\therefore \sum_{n=0}^{\infty} r^n = \frac{1}{1-r}$$

$$\therefore H(z) = \sum_{i=1}^M \frac{A_i}{1 - e^{p_i T} z^{-1}}$$

⑥

Mapping of Poles

Comparing eqⁿ ① and eqⁿ ⑥, we can conclude the transfer function of digital filter is obtained from transfer function of analog filter by performing transformation as :

$$\frac{1}{s - p_i} \longrightarrow \frac{1}{1 - e^{p_i T} z^{-1}} = \frac{z}{z - e^{p_i T}}$$

This implies that the poles from analog domain are transferred into digital domain. This transformation is called mapping of poles.

Example 2: Consider a causal Continuous-time system impulse response $h_c(t)$ and system function.

$$\textcircled{i} H_c(s) = \frac{s+a}{(s+a)^2 + b^2} \quad \text{Use IIM method}$$

determine $H(z)$ for a discrete time system such that $h_d[n] = h_c[nT]$.

→ Solution,

$$H_c(s) = \frac{s+a}{(s+a)^2 + b^2} = \frac{s+a}{(s+a+jb)(s+a-jb)} = \frac{A}{(s+a+jb)} + \frac{B}{(s+a-jb)}$$

$$A = \frac{s+a}{(s+a+jb)(s+a-jb)} \times (s+a+jb) \Big|_{s=-(a+jb)}$$

$$= \frac{-a-jb+a}{-a-jb+a-jb} = \frac{-jb}{-2jb} = 0.5$$

$$B = \frac{(s+a)}{(s+a+jb)(s+a-jb)} \times (s+a-jb) \Big|_{s=-a+jb}$$

$$= \frac{-a+jb+a}{-a+jb+a+jb} = \frac{jb}{j2b} = 0.5$$

$$H_a(s) = \frac{0.5}{(s+a+jb)} + \frac{0.5}{(s+a-jb)}$$

↓
ILT

$$h_c(t) = 0.5 e^{-i(a+jb)t} u(t) + 0.5 e^{-(a-jb)t} u(t)$$

$$= 0.5 [e^{-(a+jb)t} + e^{-(a-jb)t}] u(t)$$

Now,

$$h_1[n] = h_c[nT] = 0.5 [e^{-(a+jb)nT} + e^{-(a-jb)nT}] u[n]$$

$$H(z) = \sum_{n=0}^{\infty} h_1[n] z^{-n} = \sum_{n=0}^{\infty} 0.5 [e^{-(a+jb)nT} + e^{-(a-jb)nT}] z^{-n}$$

$$= \sum_{n=0}^{\infty} 0.5 e^{-(a+jb)nT} z^{-n} + \sum_{n=0}^{\infty} 0.5 e^{-(a-jb)nT} z^{-n}$$

$$= 0.5 \sum_{n=0}^{\infty} (e^{-(a+jb)T} z^{-1})^n + 0.5 \sum_{n=0}^{\infty} (e^{-(a-jb)T} z^{-1})^n$$

$$= 0.5 \star \frac{1}{1 - e^{-(a+jb)T} Z^{-1}} + 0.5 \star \frac{1}{1 - e^{-(a-jb)T} Z^{-1}}$$

$$= 0.5 \frac{1 - e^{-(a-jb)T} Z^{-1} + 1 - e^{-(a+jb)T} Z^{-1}}{[1 - e^{-(a+jb)T} Z^{-1}][1 - e^{-(a-jb)T} Z^{-1}]}$$

$$= 0.5 \frac{2 - [e^{-(a-jb)T} + e^{-(a+jb)T}] Z^{-1}}{1 - e^{-(a-jb)T} Z^{-1} - e^{-(a+jb)T} Z^{-1} + e^{-aT} e^{-jbT} e^{-aT} e^{jbT} Z^{-2}}$$

$$= 0.5 \frac{2 - (e^{-aT} e^{jbT} + e^{-aT} e^{-jbT}) Z^{-1}}{1 - e^{-aT} e^{jbT} Z^{-1} - e^{-aT} e^{-jbT} Z^{-1} + e^{-2aT} Z^{-2}}$$

$$= 0.5 \frac{2 - e^{-aT} (e^{jbT} + e^{-jbT}) Z^{-1}}{1 - e^{-aT} (e^{jbT} + e^{-jbT}) Z^{-1} + e^{-2aT} Z^{-2}}$$

$$= 0.5 \frac{2 - 2e^{-aT} \cos bT Z^{-1}}{1 - 2e^{-aT} \cos bT Z^{-1} + e^{-2aT} Z^{-2}}$$

$$= \frac{1 - e^{-aT} \cos bT Z^{-1}}{1 - 2e^{-aT} \cos bT Z^{-1} + e^{-2aT} Z^{-2}}$$

Hence,

$$\frac{(s+a)}{(s+a)^2 + b^2} \Rightarrow \frac{1 - e^{-aT} \cos(bT) Z^{-1}}{1 - 2e^{-aT} \cos bT Z^{-1} + e^{-2aT} Z^{-2}}$$

#

Example 3 : Repeat Example 2, if

$$H_c(s) = \frac{b}{(s+a)^2 + b^2}$$

Given that,

$$H_c(s) = \frac{b}{(s+a)^2 + b^2}$$

$$= \frac{b}{(s+a+jb)(s+a-jb)} = \frac{A_1}{(s+a+jb)} + \frac{A_2}{(s+a-jb)}$$

Where,

$$A_1 = \frac{b}{(s+a+jb)(s+a-jb)} \times (s+a+jb) \Big|_{s=-(a+jb)}$$

$$= \frac{b}{-\cancel{a}-jb + \cancel{a}-jb} = \frac{b}{-2jb} = -\frac{1}{2j}$$

Similarly,

$$A_2 = \frac{b}{(s+a+jb)(s+a-jb)} \times (s+a-jb) \Big|_{s=-(a-jb)}$$

$$= \frac{b}{-\cancel{a}+jb + \cancel{a}+jb} = \frac{1}{2j}$$

Then,

$$H_c(s) = -\frac{1/2j}{(s+a+jb)} + \frac{1/2j}{(s+a-jb)}$$

$$= \frac{1}{2j} \left[\frac{1}{s+a-jb} - \frac{1}{s+a+jb} \right] \xrightarrow{\text{ILT (Inverse Laplace Transform)}}$$

$$h_c(t) = \left[-\frac{1}{2j} e^{-(a+jb)t} + \frac{1}{2j} e^{-(a-jb)t} \right] u(t)$$

$$\text{Now, } h_1[n] = h_c[nT] = \left\{ -\frac{1}{2j} e^{-(a+jb)nT} + \frac{1}{2j} e^{-(a-jb)nT} \right\} u[n]$$

We know that,

$$H(z) = \sum_{n=0}^{\infty} h_1[n] z^{-n}$$

$$= \sum_{n=0}^{\infty} \left\{ -\frac{1}{2j} e^{-(a+jb)nT} + \frac{1}{2j} e^{-(a-jb)nT} \right\} z^{-n}$$

$$\text{or, } H_1(z) = \sum_{n=0}^{\infty} -\frac{1}{2j} e^{-(a+jb)nT} z^{-n} + \sum_{n=0}^{\infty} \frac{1}{2j} e^{-(a-jb)nT} z^{-n}$$

$$= -\frac{1}{2j} \sum_{n=0}^{\infty} \left\{ e^{-(a+jb)T} z^{-1} \right\}^n + \frac{1}{2j} \sum_{n=0}^{\infty} \left\{ e^{-(a-jb)T} z^{-1} \right\}^n$$

$$= -\frac{1}{2j} \times \frac{1}{1 - e^{-(a+jb)T} z^{-1}} + \frac{1}{2j} \times \frac{1}{1 - e^{-(a-jb)T} z^{-1}}$$

$$= \frac{1}{2j} \left\{ \frac{1}{1 - e^{-(a-jb)T} z^{-1}} - \frac{1}{1 - e^{-(a+jb)T} z^{-1}} \right\}$$

$$= \frac{1}{2j} \times \frac{1 - e^{-aT} \cdot e^{-jbT} z^{-1} - 1 + e^{-aT} \cdot e^{+jbT} z^{-1}}{1 - 2e^{-aT} (\cos bT) z^{-1} + e^{-2aT} z^{-2}} \quad \rightarrow (\text{As before})$$

$$= \frac{1}{2j} \frac{e^{-aT} (e^{jbT} - e^{-jbT}) z^{-1}}{1 - 2e^{-aT} (\cos bT) z^{-1} + 2e^{-aT} z^{-2}}$$

$$\therefore H(z) = \frac{e^{-aT} [\sin(bT)] z^{-1}}{1 - 2e^{-aT} (\cos bT) z^{-1} + 2e^{-aT} z^{-2}} \quad \text{--- (8)}$$

Hence,

$$\frac{b}{(s+a)^2 + b^2} \xrightarrow{\text{Mapping}} \frac{e^{-aT} \sin(bT) z^{-1}}{1 - 2e^{-aT} (\cos bT) z^{-1} + 2e^{-aT} z^{-2}}$$

Summary of IIM methods

→ Some standard formulae for transformation in IIM-method are as follows :

$$(i) \frac{1}{s - p_i} \xrightarrow{\text{MAPPING}} \frac{1}{1 - e^{p_i T} z^{-1}} \quad (\text{FORMULA-1})$$

$$(ii) \frac{s + a}{(s + a)^2 + b^2} \xrightarrow{\text{MAPPING}} \frac{1 - e^{-aT} \cos bT z^{-1}}{1 - 2e^{-aT} \cos bT z^{-1} + e^{-2aT} z^{-2}} \quad (\text{FORMULA-2})$$

$$(iii) \frac{b}{(s + a)^2 + b^2} \xrightarrow{\text{MAPPING}} \frac{e^{-aT} \sin bT z^{-1}}{1 - 2e^{-aT} \cos bT z^{-1} + e^{-2aT} z^{-2}} \quad (\text{FORMULA-3})$$

Note : While calculating, calculator must be in 'Radian mode'.

Relationship of s-plane to z-plane

→ Poles of the analog filters are located at, $s = p_i$
From eqn (8), We can say that the poles of digital filter, $H(z)$ are located at,

$$z = e^{p_i T} \quad \text{--- (9)}$$

Which indicates that the poles of analog filter at $s = p_i$ are transformed into the poles of digital filter at $z = e^{p_i T}$. Thus, the relationship between Laplace domain (s-domain) and z-domain is given by

$$z = e^{sT} \quad \text{--- (10)}$$

But, 's' is the Laplace operator and is expressed as,

$$s = \sigma + j\Omega$$

Where, σ is the attenuation factor
 Ω is the analog frequency.

As, In polar form, $z = re^{j\omega}$ (radius of circle)

Where 'r' is magnitude, and
' ω ' is the digital frequency.

Then, eqn (10) becomes,

$$re^{j\omega} = e^{(\sigma + j\Omega)T} = e^{\sigma T} \cdot e^{j\Omega T}$$

Comparing Real & Imaginary parts. We get

$$r = e^{\sigma T} \quad \text{--- (11) and}$$

$$\boxed{\omega = \Omega T} \quad \text{--- (12)}$$

Basically, plot in s-domain means that,

σ is plotted in x-axis and
 $j\omega$ is plotted in y-axis.

Similarly, z-domain representation means,

Real-z is plotted in x-axis and
Imaginary-z is plotted in y-axis.

From eqn (1),

$$r = e^{\sigma T}$$

Case I: $\sigma < 0 \Rightarrow 0 < r < 1$

Here, $\sigma < 0$ implies L.H.S of s-plane and
 $0 < r < 1$ implies the interior part of unit circle. Hence
L.H.S of s-plane is mapped inside the unit circle.

Case II: $\sigma = 0 \Rightarrow r = 1$

Here, $\sigma = 0$ indicates $j\omega$ axis and
 $r = 1$ indicates unit circle. Hence, $j\omega$ axis in s-plane is mapped on the unit circle.

Case III: $\sigma > 0 \Rightarrow r > 1$

Here, $\sigma > 0$ indicates R.H.S of s-plane and $r > 1$ indicates exterior of unit circle. Thus, R.H.S of s-plane is mapped outside the unit circle.

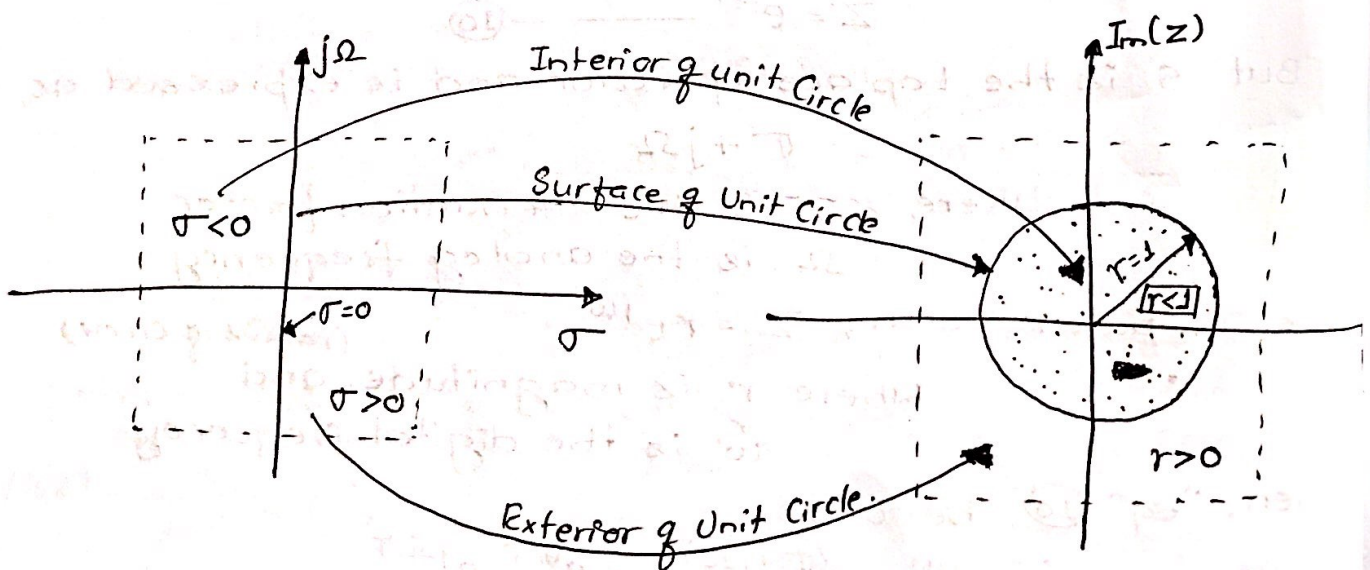


Fig: Mapping of poles from s-plane to z-plane.

Disadvantages of IIM-method

- ① Mapping from analog frequency ' Ω ' to digital frequency ' ω ' is termed as many to one (cause of aliasing effect).
- ② There will be aliasing due to the sampling process because analog filters are not bandlimited.
- ③ Change in the value of sampling interval (T) has no effect on the amount of aliasing.

ii) Bilinear Transformation Technique

→ The IIM-method has a severe limitation that they are highly suitable for LPF and certain BPF. The BLT method is based on a simple numeric integration used to derive a digital filter function from an analog filter function.

The main difference between them is that in the BLT method, the whole analog frequency range $-\infty < \Omega < \infty$ is squeezed into the unit circle $-\pi \leq \omega \leq \pi$, whereas in IIM-method the analog frequency response is folded around the unit circle indefinitely. The main advantage of the BLT method is that aliasing is avoided.

In BLT-method, the left-side of the s -plane is mapped to the inside of the unit circle in z -plane, and right side of the s -plane is mapped to the outside of unit circle in z -plane.

Let us consider the system function of an analog filter as,

$$H(s) = \frac{b}{s+a} \quad \text{————— (A)}$$

$$\text{or, } H(s) = \frac{Y(s)}{X(s)} = \frac{b}{s+a}$$

$$\text{or, } (s+a)Y(s) = bX(s)$$

$$\text{or, } sY(s) + aY(s) = bX(s)$$

Taking Inverse Laplace Transform,

$$\frac{d}{dt}y(t) + ay(t) = bx(t) \quad \text{————— (1)}$$

Instead of taking the difference equation for differential derivatives, suppose that we integrate the derivative and

(1)

approximate the integral by the trapezoidal formula, π

$$y(t) = \int_{t_0}^{t_1} y'(\tau) d\tau + y(t_0)$$

Where, $y'(t)$ = derivative of $y(t)$

$y(t_0)$ = initial value

Now, at $t_1 = nT$ and $t_0 = (n-1)T$ gives,

$$y(nT) = \frac{T}{2} [y'(nT) + y'[(n-1)T]] + y[(n-1)T] \quad \text{--- (1)}$$

from eqn (1),

$$\frac{d}{dt} y(t) = -ay(t) + bx(t)$$

$$\text{or, } y'(t) = -ay(t) + bx(t)$$

Whose, discrete equivalent value is,

$$y'(nT) = -ay[nT] + bx[nT] \quad \text{--- (2)}$$

from eqn (2) and (3), we get,

$$y(nT) = \frac{T}{2} [-ay[nT] + bx[nT] + ay[(n-1)T] + bx[(n-1)T]] + y[(n-1)T]$$

$$\text{or, } y(nT) + \frac{aT}{2} y(nT) + \frac{aT}{2} y[(n-1)T] - y[(n-1)T] = \frac{Tb}{2} x[nT] + \frac{bT}{2} x[(n-1)T]$$

$$\text{or, } y(nT) \left(1 + \frac{aT}{2}\right) - y[(n-1)T] \left(1 - \frac{aT}{2}\right) = \frac{bT}{2} \{x(nT) + x[(n-1)T]\}$$

$\updownarrow$
 Z-Transform

$$\left(1 + \frac{aT}{2}\right) Y(z) - \left(1 - \frac{aT}{2}\right) Z^{-1} Y(z) = \frac{bT}{2} \{X(z) + Z^{-1} X(z)\}$$

$$\text{or, } Y(z) \left\{ \left(1 + \frac{aT}{2}\right) - \left(1 - \frac{aT}{2}\right) Z^{-1} \right\} = \frac{bT}{2} X(z) (1 + Z^{-1})$$

$$\begin{aligned} \text{or, } H(z) = \frac{Y(z)}{X(z)} &= \frac{\frac{bT}{2} (1 + Z^{-1})}{\left(1 + \frac{aT}{2}\right) - \left(1 - \frac{aT}{2}\right) Z^{-1}} \\ &= \frac{bT/2 (1 + Z^{-1})}{(1 - Z^{-1}) + \frac{aT}{2} (1 + Z^{-1})} \end{aligned}$$

$$\frac{\frac{bT}{2} (1 + Z^{-1})}{\frac{T}{2} (1 + Z^{-1}) \left[\frac{1 - Z^{-1}}{1 + Z^{-1}} + a \right]} = \frac{b}{\frac{1 - Z^{-1}}{1 + Z^{-1}} + a}$$

$$\text{or, } H(z) = \frac{b}{\frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right) + a} \quad \text{--- (4)}$$

Comparing eqⁿ (A) and eqⁿ (4), we get,

$$S = \frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}} \quad \text{--- (5)}$$

$$S = \frac{2}{T} \frac{z-1}{z+1} \quad \text{--- (6)}$$

But, $z = re^{j\omega}$ makes eqⁿ (6) as,

$$S = \frac{2}{T} \left(\frac{re^{j\omega} - 1}{re^{j\omega} + 1} \right) = \frac{2}{T} \frac{r(\cos\omega + j\sin\omega) - 1}{(r\cos\omega + j\sin\omega) + 1}$$

$$= \frac{2}{T} \frac{(r\cos\omega - 1) + j r \sin\omega}{(r\cos\omega + 1) + j r \sin\omega}$$

$$= \frac{2}{T} \left[\frac{(r\cos\omega - 1) + j r \sin\omega}{(r\cos\omega + 1) + j r \sin\omega} \times \frac{(r\cos\omega + 1) - j r \sin\omega}{(r\cos\omega + 1) + j r \sin\omega} \right]$$

$$= \frac{2}{T} \left[\frac{r^2 \cos^2\omega - 1 + r^2 \sin^2\omega + 2j r \sin\omega}{(r\cos\omega + 1)^2 + r^2 \sin^2\omega} \right]$$

$$= \frac{2}{T} \times \frac{r^2(\cos^2\omega + \sin^2\omega) - 1 + 2j r \sin\omega}{r^2 \cos^2\omega + 2r\cos\omega + 1 + r^2 \sin^2\omega}$$

$$= \frac{2}{T} \left[\frac{r^2 - 1 + 2j r \sin\omega}{r^2(\cos^2\omega + \sin^2\omega) + 1 + 2r\cos\omega} \right]$$

$$S = \frac{2}{T} \left[\frac{r^2 - 1 + j 2r \sin\omega}{r^2 + 1 + 2r\cos\omega} \right]$$

$$= \frac{2}{T} \left[\frac{r^2 - 1}{r^2 + 1 + 2r\cos\omega} + j \frac{2r \sin\omega}{r^2 + 1 + 2r\cos\omega} \right]$$

$$= \sigma + j\omega$$

Now, Comparing Real parts and Imaginary parts, We get

$$\sigma = \frac{2}{T} \left(\frac{r^2 - 1}{r^2 + 1 + 2r \cos w} \right) \text{ and, } \text{--- (7)}$$

$$\Omega = \frac{2}{T} \left[\frac{2r \sin w}{r^2 + 1 + 2r \cos w} \right] \text{ --- (8)}$$

Here, eqⁿ (7) implies,

case I: If $r \leq 1$ then $\sigma < 0$

case II: If $r > 1$ then $\sigma > 0$

i.e L.H.S of s-plane is mapped inside the unit circle in z-plane and R.H.S of s-plane is mapped outside the unit circle in z-plane. When, $r = 1$, $\sigma = 0$ i.e on the unit circle. So, For $r = 1$, eqⁿ (8) becomes,

$$\Omega = \frac{2}{T} \times \frac{2 \sin w}{1 + 1 + 2 \cos w}$$

$$= \frac{2}{T} \times \frac{2 \sin w}{2 + 2 \cos w}$$

$$= \frac{2}{T} \times \frac{2 \sin w}{2(1 + \cos w)}$$

$$= \frac{2}{T} \times \frac{2 \sin(w/2) \cos(w/2)}{2 \cos^2 w/2}$$

$$= \frac{2}{T} \frac{\sin(w/2)}{\cos(w/2)}$$

$$\therefore \boxed{\Omega = \frac{2}{T} \tan\left(\frac{w}{2}\right)} \text{ is Analog frequency.} \text{--- (9)}$$

or, the frequency in digital domain is given by,

$$\boxed{w = 2 \tan^{-1} \left(\frac{\Omega T}{2} \right)} \text{ --- (10)}$$

The relationship in eqⁿ (10) is plotted as :

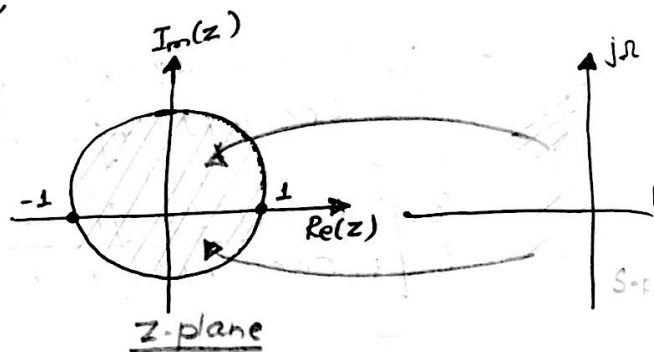


Fig. BLT mapping

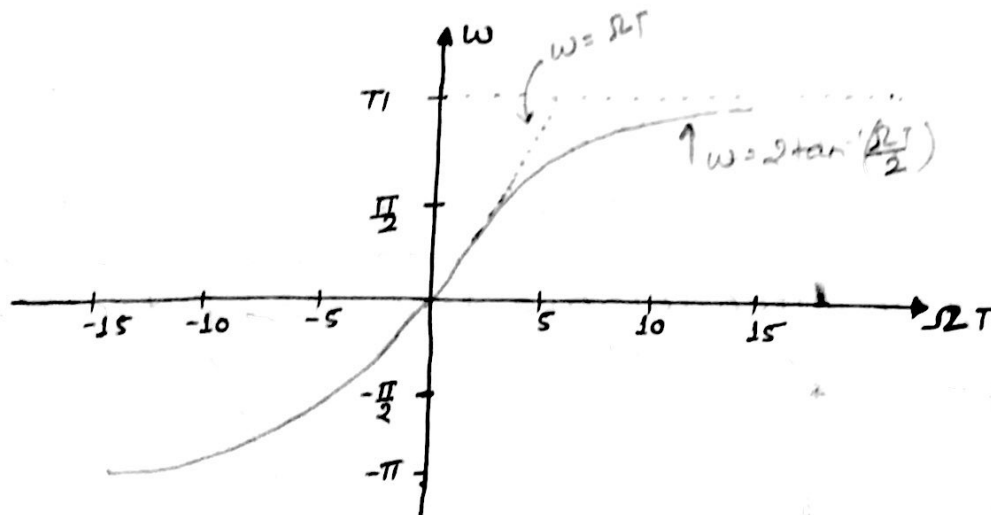


Fig: Mapping of the analog Frequency ' Ω ' to the Corresponding digital Frequency ' ω ' resulting from the bilinear Transform

Conclusion

- ① It is clear from the plot that positive imaginary axis ($j\Omega$) in s-plane is mapped into the upper-half of the unit Circle in the z-plane and negative imaginary axis ($-j\Omega$) in the s-plane is mapped into lower half of the z-plane
- ② The mapping is highly Non-linear.
 - (a) The complete negative axis in s-plane from $\Omega = -\infty$ to 0 is mapped to the lower half of the unit Circle from $\omega = -\pi$ to $\omega = 0$ or $z = -1$ to $z = +1$.
 - (b) The Complete positive axis in s-plane from $\Omega = 0$ to $\Omega = \infty$ is mapped to the upper half of the unit Circle from $\omega = 0$ to $\omega = \pi$ or $z = +1$ to $z = -1$.

The non-linearity in the relation between ' Ω ' and ' ω ' is quite often referred to as "frequency WARPING" or "frequency Compression".

WARPING EFFECT

→ The mapping of frequency from ' Ω ' to ' ω ' is approximately linear for small value of ' Ω ' and ' ω ', for the higher frequency however the relation between ' Ω ' and ' ω ' becomes highly non-linear. This introduces the distortion in the frequency scale of digital filter relative to analog filter. This effect is known as warping effect.

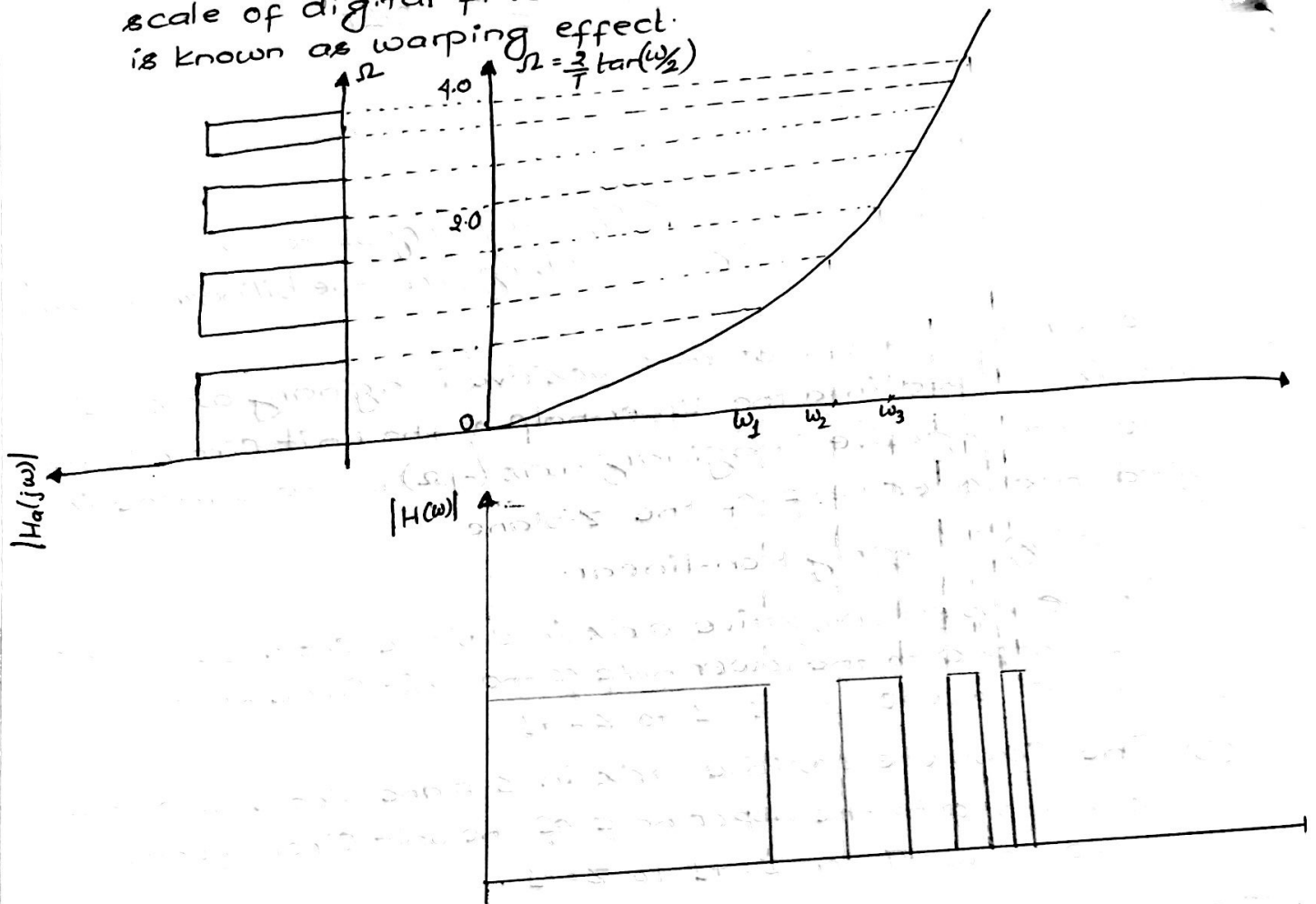


Fig: Influence of Warping effect on amplitude resp

The influence of warping effect on the amplitude response can be demonstrated by considering an analog filter with number of passband centred at regular desired digital filter. The desired digital filter has same number of passband but the centre frequencies and the bandwidth of higher frequency passbands in digital domain tends to be reduced.

Advantages of BLT-method

- (i) There is one-to-one mapping from the s-domain to z-domain.
- (ii) There is no aliasing effect.
- (iii) Stable ~~pole~~ analog filter is transformed into the stable digital filter.

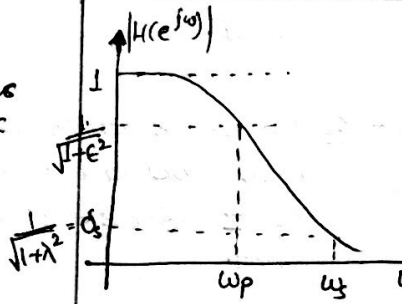
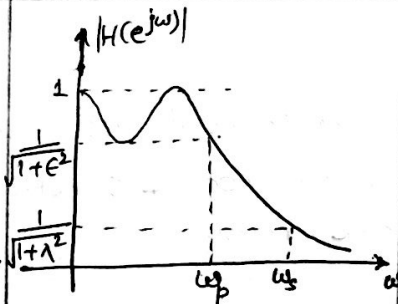
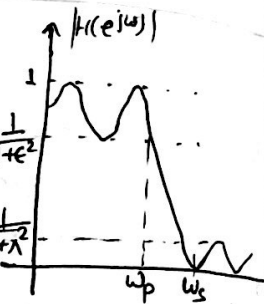
Difference between Analog filter & Digital Filter

Analog filter	Digital filter
① It works on analog signals or actual signal.	① It operates on the digital samples of the signal.
② It is defined by linear differential equations.	② They are defined using linear difference equation.
③ Electrical components like resistors, inductors, capacitors are used	③ adders, subtractors, delays or other logical components are used.
④ To be stable and causal, $H(s)$ must be a rational function of s , whose coefficients are real.	④ To be stable & causal, $H(z)$ must be a rational function of z , whose coefficients are real.
⑤ For stability & causality, the poles of transfer function should lie on the left half of s -plane.	⑤ For stability and causality the poles of transfer function should lie inside the unit circle in z -plane.

Difference between IIM and BLT method.

Impulse Invariance Method	Bilinear Transformation method.
poles are transferred by using the expression : $\frac{1}{s-p_i} \rightarrow \frac{1}{1-e^{p_i T} z^{-1}}$	① poles are transferred as $s = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right)$
Mapping is many to one	② Mapping is one to one.
Aliasing effect is present	③ No aliasing effect.
It is not suitable to design HPF & BSF	④ suitable to design all-types of filter.
Only poles of the system can be mapped.	⑤ poles as well as zeroes can be mapped.
No frequency warping effect	⑥ Frequency warping is present.

Comparison of different IIR filter design method

Parameters	Butterworth	Chebyshev (Type 2)	Elliptic
Ripple in Passband	No	Yes	Yes
Ripple in Stopband	No	No	Yes
Transition BW for fixed order	Widest	Narrower	Narrowest
Order for given filter specification	Highest	Lower	Lowest
Magnitude Characteristics Curve for LPF			
Order of filter (N)	$\frac{\log(\lambda/\epsilon)}{2 \log(\frac{\Omega_s}{\Omega_p})}$ $\lambda = \sqrt{10^{0.1\alpha_s-1}}$ $\epsilon = \sqrt{10^{0.1\alpha_p-1}}$	$\frac{\cosh^{-1}(\lambda/\epsilon)}{\cosh^{-1}(\frac{\Omega_s}{\Omega_p})}$	$\frac{K(\frac{\Omega_p}{\Omega_s}) K \sqrt{1 - \frac{\epsilon^2}{\delta^2}}}{K(\frac{\epsilon}{\delta}) K \sqrt{1 - (\frac{\Omega_p}{\Omega_s})^2}}$ $\delta_2 = 10 \log \sqrt{1 + \epsilon^2} - \sqrt{1 + \delta^2}$

Advantage/Disadvantages of digital filter

Advantages

1. It has better Accuracy
2. It is free of component tolerances
3. It can be implemented in hardware/software
4. Repeatable performance.
5. flexibility of configuration
6. Easily portable.

Disadvantages

1. Quantization noise is present
2. Not suitable of simple filters & low cost systems.
3. Finite register length affect the performance.
4. Analog data can be processed directly.

Difference between FIR and IIR filters

parameters	Finite Impulse Response filter	Infinite Impulse Response filter.
pole-zero location	All zero system	pole-zero system or All-pole filter.
Transfer function	$H(z) = \sum_{k=0}^M b_k z^{-k}$	$H(z) = \frac{\sum_{k=0}^M b_k z^{-k}}{1 + \sum_{k=1}^N a_k z^{-k}}$
Efficiency	→ low	→ High
Speed	→ Slow	→ slow Fast
stability	→ Always stable	→ Conditionally stable
phase Response	→ Linear	→ Non-linear
Limit cycles	→ Doesnot produce	→ may produce.
Computational Complexity	→ More operations	→ fewer operations.

ii) Matched Z-Transform Method

→ It is one of the very basic method to Convert an analog domain into discrete domain. This method directly maps the poles and zeroes of an analog filter directly into poles and zeroes of the z-plane in one-to-one manner.

Suppose that the system function of the analog filter is expressed in the factored form :

$$H(s) = \frac{\prod_{k=1}^M (s - z_k)}{\prod_{k=1}^N (s - p_k)}$$

Where, z_k and p_k are the poles of the analog filter for converting the above analog transfer function into digital filter. Apply 'z' transform to the above equation,

$$H(z) = \frac{\prod_{k=1}^M (1 - e^{z_k T} z^{-1})}{\prod_{k=1}^N (1 - e^{p_k T} z^{-1})}$$

Where, T is the sampling interval.

The above mapping Technique is called as a Matched-z transform technique. To preserve the frequency characteristics of the analog filter, the sampling interval (T) in the matched-z transformation must be properly selected to yield the poles and zeroes location at the equivalent position in the "Z" plane.

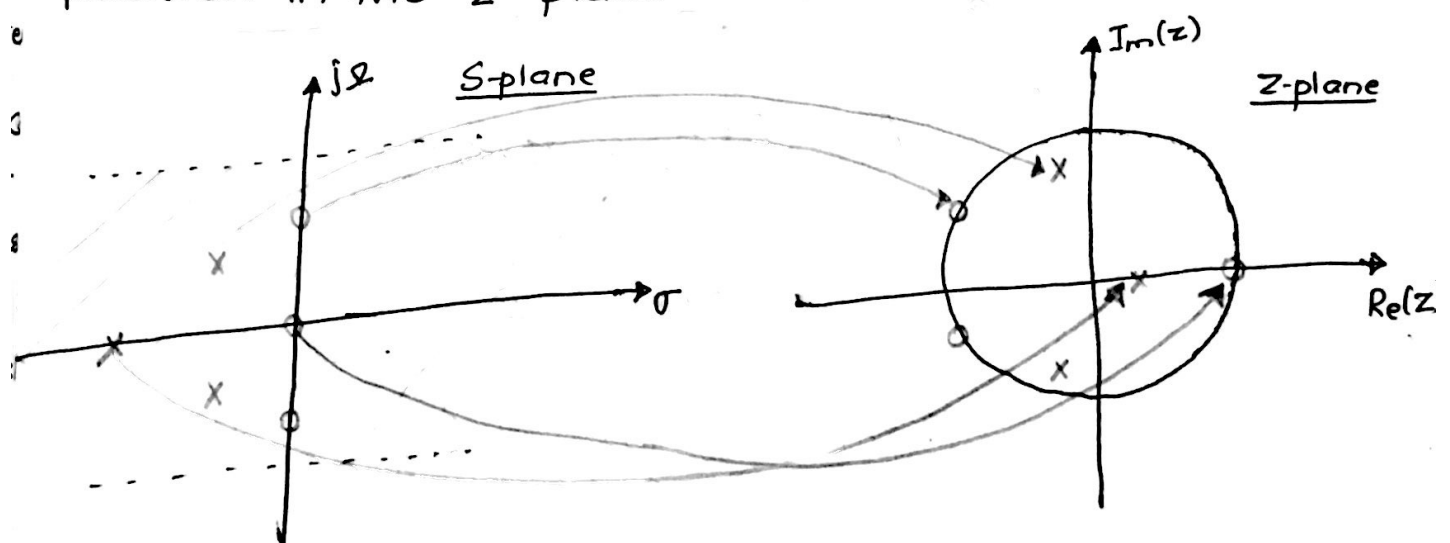


Fig: Mapping of poles/zeroes of s-plane to z-plane

Ex: (DAS MAT COMPUTER)

Conclusion

↳ poles and zeroes are transformed according to
 $z_k \rightarrow e^{z_k T}$ and $p_k \rightarrow e^{p_k T}$

↳ poles using this method are similar to IIM-meth, but zeroes are located at new position.

Advantage

- (i) It is easy to implement.
- (ii) We can avoid the aliasing effect.

Disadvantage

- (i) If the analog system has zeroes with center frequencies greater than half of the sampling frequency, the matched-z transformation is unsuitable.

Example 4: Consider the following analog filter into a digital IIR filter using IIM method and matched-z transformation method and compare the result.

$$H_a(s) = \frac{s+2}{(s+1)(s+3)}$$

→ Given that,

$$H_a(s) = \frac{s+2}{(s+1)(s+3)} \quad \text{--- (1)}$$

Case I:

IIM method

$$H_a(s) = \frac{s+2}{(s+1)(s+3)} = \frac{A_1}{s+1} + \frac{A_2}{s+3}$$

$$\begin{aligned} \text{Where, } A_1 &= \frac{s+2}{(s+1)(s+3)} \times (s+1) \Big|_{s=-1} \\ &= \frac{(-1+2)}{(-1+3)} = \frac{1}{2} \end{aligned}$$

$$\begin{aligned} A_2 &= \frac{s+2}{(s+1)(s+3)} \times (s+3) \Big|_{s=-3} \\ &= \frac{-3+2}{-3+1} = -\frac{1}{2} \end{aligned}$$

So,

$$H_a(s) = \frac{1}{2} \times \frac{1}{s+1} + \frac{1}{2} \times \frac{1}{s+3}$$

Using IIM-method formula,

$$\begin{aligned}
 H(z) &= \frac{1}{2} \times \frac{1}{1-e^{-T}z^{-1}} + \frac{1}{2} \times \frac{1}{1-e^{-3T}z^{-1}} \\
 &= \frac{1}{2} \left[\frac{1}{1-e^{-T}z^{-1}} + \frac{1}{1-e^{-3T}z^{-1}} \right] \\
 &= \frac{1}{2} \left[\frac{(1-e^{-3T}z^{-1}) + (1-e^{-T}z^{-1})}{(1-e^{-T}z^{-1})(1-e^{-3T}z^{-1})} \right] \\
 &= \frac{1}{2} \left[\frac{2 - (e^{-T} + e^{-3T})z^{-1}}{(1-e^{-T}z^{-1})(1-e^{-3T}z^{-1})} \right] \\
 &= \frac{1 - \frac{1}{2}(e^{-T} + e^{-3T})z^{-1}}{(1-e^{-T}z^{-1})(1-e^{-3T}z^{-1})} \\
 &= \frac{z \left\{ z - \frac{1}{2}(e^{-T} + e^{-3T}) \right\}}{z^2 \{(z - e^{-T})(z - e^{-3T})\}} \\
 &= \frac{z \left\{ z - \frac{1}{2}(e^{-T} + e^{-3T}) \right\}}{(z - e^{-T})(z - e^{-3T})}
 \end{aligned}$$

Therefore, Using IIM method

Zeros location are at :

$$\begin{cases} z = 0 & \& \\ z = \frac{1}{2}(e^{-T} + e^{-3T}) \end{cases}$$

poles location are at :

$$\begin{cases} z = e^{-T} \text{ and} \\ z = e^{-3T} \end{cases}$$

Case II :

Matched-Z Transformation method

eqn ① becomes,

$$H(z) = \frac{1 - e^{-2T}z^{-1}}{(1 - e^{-T}z^{-1})(1 - e^{-3T}z^{-1})}$$

$$\text{Or, } H(z) = \frac{1 - e^{-2T} z^{-1}}{(1 - e^{-T} z^{-1})(1 - e^{-3T} z^{-1})}$$

$$= \frac{z(z - e^{-2T})}{(z - e^{-T})(z - e^{-3T})}$$

Therefore,

Zeros are located at,

$$\begin{cases} z_1 = 0 & \& \\ z_2 = e^{-2T} \end{cases}$$

poles are located at,

$$\begin{cases} p_1 = e^{-T} & \& \\ p_2 = e^{-3T} \end{cases}$$

This proves that location of poles using IIM, and matched z-transform methods are same w^h location of zeros are different.

Example 5: Use the matched-z-transformation to convert analog filter with system function,

$$H(s) = \frac{s + 0.1}{(s + 0.1)^2 + 9} \text{ into a digital IIR}$$

filter. Select, $T = 0.1$ and compare the location of z in $H(z)$ with the locations of the zeros obtained by applying the IIM-method in the conversion of $H(s)$

→ Given that,

$$H(s) = \frac{s + 0.1}{(s + 0.1)^2 + 9}$$

case I: Matched z-transformation method.

$$H(s) = \frac{s + 0.1}{(s + 0.1)^2 + (3)^2}$$

$$= \frac{s + 0.1}{(s + 0.1 + j3)(s + 0.1 - j3)}$$

$$= \frac{s + 0.1}{[s + (0.1 + j3)][s + (0.1 - j3)]}$$

Using Matched-Z-transformation method,

$$H(z) = \frac{1 - e^{-0.1T} z^{-1}}{[1 - e^{-(0.1+j3)T} z^{-1}][1 - e^{-(0.1-j3)T} z^{-1}]}$$

$$= \frac{z(z - e^{-0.1T})}{[z - e^{-(0.1+j3)T}][z - e^{-(0.1-j3)T}]}$$

as $T = 0.1 \text{ sec}$

$$= \frac{z(z - e^{-0.1 \times 0.1})}{[z - e^{-(0.1+j3)0.1}][z - e^{-(0.1-j3)0.1}]}$$

$$= \frac{z(z - 0.99)}{(z - e^{-0.1 \times 0.1} \cdot e^{-j3 \times 0.1})(z - e^{-0.1 \times 0.1} \cdot e^{+j3 \times 0.1})}$$

$$H(z) = \frac{z(z - 0.99)}{(z - 0.99e^{-j0.3})(z - 0.99e^{j0.3})} = \frac{z(z - 0.99)}{z^2 - 0.99e^{j0.3}z - 0.99e^{-j0.3}z + (0.99)^2}$$

$$= \frac{z(z - 0.99)}{z^2 - 0.99z(e^{j0.3} + e^{-j0.3}) + 0.98}$$

$$= \frac{z(z - 0.99)}{z^2 - 2 \times 0.99 \times \cos(0.3)z + 0.98}$$

$$= \frac{z(z - 0.99)}{z^2 - 1.89z + 0.98}$$

Hence,

zeros location,

$$\left. \begin{aligned} z_1 &= 0 \\ \text{and, } z_2 &= 0.99 \end{aligned} \right\}$$

poles location,

$$\left. \begin{aligned} p_1 &= 0.99e^{j0.3} = 0.945 + j0.29 \\ \text{and, } p_2 &= 0.99e^{-j0.3} = 0.945 - j0.29 \end{aligned} \right\}$$

Case II : IMPULSE INVARIANCE METHOD

$$H_a(s) = \frac{s + 0.1}{(s + 0.1)^2 + 9}$$

$$= \frac{s + 0.1}{(s + 0.1)^2 + (3)^2} \left[\text{in the form } \frac{(s+a)}{(s+a)^2 + b^2} \right]$$

Using the formula,

$$H(z) = \frac{1 - e^{-aT} \cos bT z^{-1}}{1 - 2e^{-aT} \cos bT z^{-1} + e^{-2aT} z^{-2}} \quad \left(\begin{aligned} a &= 0.1, b = 3 \\ T &= 0.1 \end{aligned} \right)$$

$$= \frac{1 - e^{-0.1 \times 0.1} \cos(3 \times 0.1) z^{-1}}{1 - 2e^{-0.1 \times 0.1} \cos(3 \times 0.1) z^{-1} + e^{-2 \times 0.1 \times 0.1} z^{-2}}$$

$$\text{or, } H(z) = \frac{1 - 0.945z^{-1}}{1 - 1.892z^{-1} + 0.98z^{-2}}$$

$$= \frac{z(z - 0.945)}{(z^2 - 1.892z + 0.98)}$$

Where, zeroes location,

$$\left. \begin{aligned} z_1 &= 0.8 \\ z_2 &= 0.945 \end{aligned} \right\}$$

poles location,

$$\left. \begin{aligned} p_1 &= -0.945 + j0.295 \\ p_2 &= -0.945 + j0.295 \end{aligned} \right\}$$

Here, In two methods,
poles location are same but,
zeroes location are different.

Example 6: Repeat same example 5: for BLT method
IIM method.

$$\rightarrow \text{Given, } H_a(s) = \frac{(s + 0.1)}{(s + 0.1)^2 + 9}$$

case I: Using BLT-method

$$H(z) = H_a(s) \Big|_{s = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right) = \frac{2}{0.1} \left(\frac{1-z^{-1}}{1+z^{-1}} \right) = 20 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)}$$

$$= 20 \frac{(z-1)}{(z+1)}$$

$$= \frac{20 \left(\frac{z-1}{z+1} \right) + 0.1}{\left(20 \frac{z-1}{z+1} + 0.1 \right)^2 + 9}$$

$$= \frac{\frac{20(z-1) + 0.1(z+1)}{(z+1)}}{\frac{[20(z-1) + 0.1(z+1)]^2 + 9(z+1)^2}{(z+1)^2}}$$

(26)

$$\begin{aligned}
 &= \frac{(20z - 20 + 0.1z + 0.1)(z+1)}{\{20(z-1)\}^2 + 2 \times 20(z-1) \times 0.1(z+1) + \{0.1(z+1)\}^2 + 9(z+1)^2} \\
 &= \frac{(20.1z - 19.9)(z+1)}{400(z^2 - 2z + 1) + 4(z^2 - 1) + 0.01(z^2 + 2z + 1) + 9(z^2 + 2z + 1)} \\
 &= \frac{20.1z^2 + 20.1z - 19.9z - 19.9}{400z^2 - 800z + 400 + 4z^2 - 4 + 0.01z^2 + 0.02z + 0.01 + 9z^2 + 18z + 9} \\
 &= \frac{20.1z^2 - 0.2z - 19.9}{413.01z^2 - 781.98 + 405.01}
 \end{aligned}$$

$$\therefore H(z) = \frac{(z-1)(z+0.99)}{[z - (-0.95 + j0.3)][z - (-0.95 - j0.3)]}$$

So, location of zeroes,
 $z_1 = 1$
 $z_2 = -0.99$

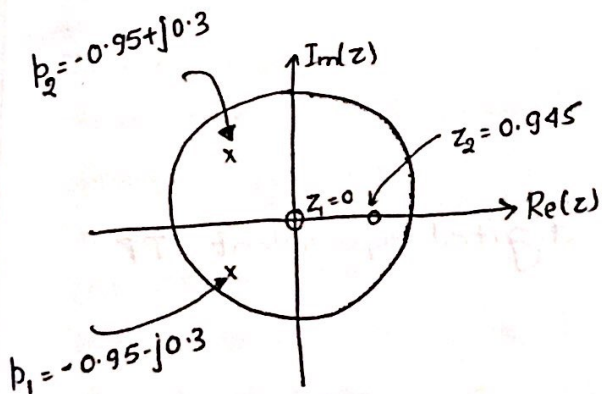
and, location of poles,
 $p_1 = -0.95 + j0.3$
 $p_2 = -0.95 - j0.3$

Case II: Using IIM method (calculated in earlier example)

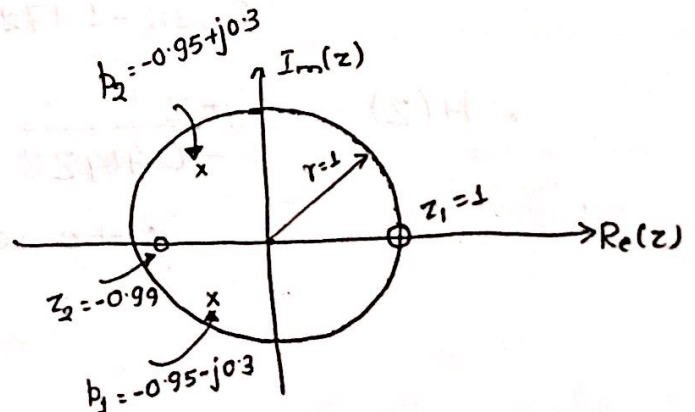
location of zeroes,
 $z_1 = 0$
 $z_2 = 0.945$

location of poles,
 $p_1 = -0.945 + j0.295 \approx -0.95 + j0.3$
 $p_2 = -0.945 - j0.295 \approx -0.95 - j0.3$

Conclusion: (i) poles location are approximately same
(ii) zeroes location are completely different.



IIM method



BLT method

Example It is required to design a digital filter with a Bandwidth of 0.25π from an analog filter having the following system response

$$H_a(s) = \frac{\Omega_c}{s + \Omega_c} \quad \text{Using bilinear transform}$$

to obtain $H(z)$.

→ Given that,

$$\text{Cut-off frequency, } \omega_c = 0.25\pi \text{ rad.}$$

Prewarping the frequency (BLT-method) we get,

$$\begin{aligned} \Omega_c &= \frac{2}{T} \tan\left(\frac{\omega_c}{2}\right) = \frac{2}{T} \tan\left(\frac{0.25 \times 180}{2}\right) \\ &= \frac{0.828}{T} \text{ rad/sec} \end{aligned}$$

Now,

Analog Transfer function,

$$H_a(s) = \frac{\Omega_c}{s + \Omega_c} \quad \text{is transformed into digital}$$

Transfer function $H(z)$ using BLT method as,

$$H(z) = H_a(s) \Big|_{s = \frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}}} \quad \& \quad \Omega_c = \frac{0.828}{T}$$

We get,

$$H(z) = \frac{\frac{0.828}{T}}{\frac{2}{T} \frac{1-z^{-1}}{1+z^{-1}} + \frac{0.828}{T}}$$

$$= \frac{0.828(1+z^{-1})}{2(1-z^{-1}) + 0.828(1+z^{-1})}$$

$$= \frac{0.828(1+z^{-1})}{2.828 - 1.172z^{-1}}$$

$$\therefore H(z) = \frac{1+z^{-1}}{1-0.414z^{-1}}$$

is the required digital equivalent

Example: Convert the analog filter with system function,
 $H_a(s) = \frac{s+0.1}{(s+0.1)^2 + 16}$ into digital IIR filter by

means of bilinear transformation. Resonant frequency of digital filter is given by $\omega_r = \pi/2$. Also determine the pole-zero location.

→ Given that,

$$H_a(s) = \frac{s+0.1}{(s+0.1)^2 + 16}$$

So, $H_a(s)$ has a resonant frequency,
 $\Omega_r = \sqrt{16} = 4$

Now, We have, $\Omega_r = \frac{2}{T} \tan\left(\frac{\omega}{2}\right)$

$$\text{or, } 4 = \frac{2}{T} \tan\left(\frac{\pi/2}{2}\right)$$

$$\text{or, } 4 = \frac{2}{T} \tan\left(\frac{\pi}{4}\right)$$

$$\therefore T = \frac{1}{2}$$

Since BLT method is used,

$$\text{map } s = \frac{2}{T} \left(\frac{1-z^{-1}}{1+z^{-1}} \right) = 4 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)$$

$$\text{Then, } H(z) = H_a(s) \Big|_{s=4 \left(\frac{1-z^{-1}}{1+z^{-1}} \right)}$$

(115)

$$\text{or, } H(z) = \frac{4 \frac{1-z^{-1}}{1+z^{-1}} + 0.01}{\left[4 \frac{1-z^{-1}}{1+z^{-1}} + 0.01\right]^2 + 16}$$

$$= \frac{0.128 + 0.006Z^{-1} + 0.122Z^{-2}}{1 + 0.0006Z^{-1} + 0.975Z^{-2}}$$

$$= \frac{0.128 + 0.006Z^{-1} - 0.122Z^{-2}}{1 + 0.975Z^{-2}}$$

$$= \frac{(z+1)(z-0.95)}{(z-0.987e^{-j\pi/2})(z-0.987e^{j\pi/2})}$$

This filter has a pole at,

$$P_{1,2} = 0.987e^{\pm j\pi/2}$$

and, zeroes at,

$$Z_{1,2} = -1, 0.95$$